Interest-Driven Personalized Sports Recommendation: A Multi-Task Learning Approach for Playing, Watching, and Professional Potential Prediction

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***Abstract*—Existing sports matching systems rely primarily on physical fitness metrics to recommend sports to users, leaving personal interests as an underutilized signal despite strong evidence from sports psychology that interest and intrinsic motivation are the strongest predictors of sustained engagement. We present an *interest-driven* sports recommendation framework that positions personal interests as the primary input signal, with physical strengths and demographic features as secondary factors. Our system simultaneously predicts sports preferences across three engagement levels—playing, watching, and professional potential—using a shared-representation multi-task neural network. We additionally introduce a *discovery score*, a cosine-similarity metric that surfaces sports unfamiliar to the user based on latent interest alignment. We evaluate on a literature-calibrated synthetic dataset of 1,000 users across 20 sports, with feature-label correlations anchored to published findings in sports psychology. Our multi-task model achieves an NDCG@5 of 0.467, representing a 10.1% improvement over a physical-only XGBoost baseline (NDCG@5 = 0.424). SHAP attribution confirms that interest features account for 35.4% of predictive importance individually, and 65.9% when combined with self-rated strength features—collectively outweighing objective physical metrics. The discovery module achieves an 85.6% novelty rate at K=5. To our knowledge, this is the first unified framework that is simultaneously interest-first, multi-level (play/watch/pro), designed for general and novice users, and equipped with a novel-sport discovery mechanism. Code and data are released at https://github.com/KusalPabasara/sports-recommendation.**

***Index Terms*—sports recommendation, multi-task learning, interest-driven, discovery score, personalized recommendation, physical activity**

I. Introduction

Sports participation is widely associated with physical and mental health benefits, yet global participation rates remain low, with the WHO reporting that over 1.4 billion adults are insufficiently active [1]. A significant contributor to this gap is the challenge of sports *discovery*—most people, particularly students and novices, are exposed to a narrow slice of the sports landscape, often through school curricula or geographic availability, with little guidance on which sports might genuinely suit their personality and interests.

Existing technology-based solutions to this problem focus almost exclusively on physical traits. Commercial systems such as AI Sport Scout [2] match users to sports via battery fitness tests (sprint times, jump distances, flexibility scores), implicitly assuming that physical aptitude is the primary driver of sports participation. While physical ability is undeniably relevant for competitive performance, a substantial body of sports psychology research establishes that *intrinsic motivation*—driven by personal interest and enjoyment—is the stronger predictor of sustained participation and long-term engagement [3]–[5].

This paper addresses the following research question: *Can a machine learning system that prioritizes personal interests over physical traits provide better and more holistic sports recommendations to general users?*

We make the following contributions:

- We propose the first **interest-first sports recommendation framework**, placing Likert-scaled personal interest profiles as the dominant input signal, with physical and demographic features as secondary factors.
- We design a **multi-task neural network** that simultaneously predicts three engagement levels—*play*, *watch*, and *professional potential*—via a shared representation backbone and task-specific output heads.
- We introduce a **discovery score**, a cosine-similarity metric that surfaces sports the user has never encountered, ranked by latent interest alignment—achieving a novelty rate of 85.6% at K=5.
- We demonstrate via SHAP attribution that interest features account for **65.9%** of total predictive importance when combined with self-rated strength features, validating the interest-first design.
- We release a **literature-calibrated synthetic dataset** of 1,000 users across 20 sports as a reproducible research

artifact, together with full code.

The remainder of this paper is structured as follows. Section II reviews related work. Section III presents the methodology. Section IV reports experimental results. Section V discusses findings, limitations, and future work. Section VI concludes.

II. Related Work

A. Sports Talent Identification and Athlete Performance Prediction

Machine learning has been widely applied to predict athlete performance and identify sporting talent. Ensemble methods such as Random Forests and XGBoost demonstrate strong predictive accuracy on physiological and biomechanical data [6], [7]. Systematic reviews of talent identification confirm that morphological and physiological variables dominate current models, with psychological factors receiving far less attention [8]. Commercial systems such as AI Sport Scout [2] operationalize this approach by matching users to sports through battery fitness tests. While effective for elite scouting, these systems require specialized testing infrastructure, presuppose physical aptitude as the primary signal, and are inaccessible to general or novice users. Critically, none model interest or motivation features, and the ceiling of physical-only classification has been quantified at approximately 60% explained variance [9], leaving the remainder unexplained.

B. Recommendation Systems and Personalized Activity Suggestions

Recommendation systems broadly fall into content-based, collaborative filtering, and hybrid approaches [10]. Collaborative filtering leverages user–item interaction history [11] but suffers from a cold-start problem that makes it unsuitable for novice users with no prior sports history. Deep learning-based recommenders, including neural collaborative filtering and multi-task architectures [12], improve expressiveness but rarely incorporate non-behavioral features such as interests or physical characteristics. Closest to our work, personalized physical activity recommendation systems have shown that tailored suggestions increase adherence by up to 23% over generic advice [13]; however, these systems use physical metrics as the primary signal and do not model watching or professional potential, nor do they guide users toward unfamiliar sports.

C. Psychological Factors in Sports Participation

Sports psychology provides compelling theoretical grounding for an interest-first approach. Vallerand et al.'s dualistic model of passion [4] establishes that harmonious passion—intrinsically motivated and aligned with personal identity—predicts sustained engagement, while obsessive passion leads to burnout. Self-determination theory [3] further confirms that intrinsic motivation, driven by interest and enjoyment, is the strongest longitudinal predictor of sport continuation. Longitudinal youth studies show that interest and enjoyment outperform physical talent in predicting sport persistence, with mismatch between assigned sports and personal interest being a primary driver of dropout [5]. Personality research additionally links openness and extraversion to sport type preference [14], suggesting that interest-adjacent psychological features carry predictive signal. Despite this evidence, ML systems in sport have not operationalized interest features as first-class inputs.

TABLE I
Comparison with Related Work

Work	Interest-first	Multi-level	Novice	Discovery
AI Sport Scout [2]	×	×	×	×
Ramos et al. [8]	×	×	×	×
Opast recs. [13]	×	×	✓	×
IARCI student [15]	partial	×	✓	×
Ma et al. MTL [18]	×	partial	×	×
Ours	✓	✓	✓	✓

D. Student and Youth Sports Engagement

At the intersection of ML and youth sports, prediction of student sport preferences has been explored using decision trees and Naive Bayes classifiers on small survey samples [15], achieving approximately 78% accuracy in classifying a single preferred sport from three to five options. However, this approach does not generalize to multi-label recommendation across a broad sports taxonomy, does not model watching or professional potential, and introduces no discovery mechanism. Qualitative evidence from physical education settings suggests that interest alignment improves participation by approximately 40% compared to teacher-assigned sports [16], motivating an automated, scalable approach.

E. Multi-Task Learning in Recommendation

Multi-task learning (MTL) [17] trains a shared representation across related prediction objectives, consistently improving generalization under limited data. In recommendation, MTL is particularly effective when tasks share latent user preferences [18], as demonstrated by Google's Multi-gate Mixture-of-Experts framework for app recommendations. We adapt this paradigm to the sports domain, where playing preference, watching preference, and professional potential share a common latent interest structure but diverge in their specific feature correlates. SHAP-based feature attribution [19] is used to verify that interest features dominate predictions across all three tasks.

F. Research Gap

Table I summarizes the gap between existing work and our contributions. No existing system simultaneously (i) places interests as the primary input signal, (ii) predicts sports preferences across play, watch, and professional levels, (iii) targets general and novice users, and (iv) surfaces *novel* sports via a discovery mechanism.

III. Methodology

A. Problem Formulation

Let a user u be represented by a feature vector $\mathbf{x}_u \in \mathbb{R}^d$ comprising interest, strength, physical, and demographic features. Let $\mathcal{S} = \{s_1, \ldots, s_N\}$ denote the sports taxonomy of $N = 20$ sports. We define three binary label vectors per user: $\mathbf{y}^{(\text{play})}, \mathbf{y}^{(\text{watch})}, \mathbf{y}^{(\text{pro})} \in \{0,1\}^N$. The goal is to learn a function $f : \mathbb{R}^d \to [0,1]^{3N}$ that jointly predicts the probability of engagement with each sport at each level.

B. Dataset Construction

No public dataset combines interest profiles, physical metrics, and multi-label sports engagement labels. We generate a literature-calibrated synthetic dataset of $M = 1{,}000$ users as a reproducible proxy, following precedents for feasibility studies in data-scarce domains.

1) Feature Schema: Each user u is described by $d = 29$ features across four groups:

- **Interests** (12 Likert 1–5 dimensions): team vs. individual preference, outdoor/indoor, competition drive, risk tolerance, creative expression, social enjoyment, endurance interest, power interest, speed/agility interest, spectator engagement, ambition level, strategy preference.
- **Strengths** (8 Likert 1–5 dimensions): self-rated endurance, strength, speed, flexibility, coordination, agility, reaction time, strategic ability.
- **Physical metrics** (6 continuous): height (cm), weight (kg), BMI (derived), 100m sprint time (s), standing jump (cm), age.
- **Demographics** (3): gender, region, facility access.

2) Literature-Calibrated Label Generation: For each user–sport pair, a raw affinity score is computed as:

$$a_{u,s} = 2 \cdot \frac{\mathbf{i}_u \cdot \mathbf{p}_s}{D_I} + 0.5 \cdot \phi_{u,s} + b_s + \epsilon \tag{1}$$

where $\mathbf{i}_u \in [0,1]^{D_I}$ is the normalized interest vector, $\mathbf{p}_s \in [0,1]^{D_I}$ is a hand-crafted sport profile vector (e.g., football: high team, high outdoor, high endurance), $\phi_{u,s} \in [0,1]$ is a physical fit score derived from strength self-ratings, b_s is the sport's global popularity prior, and $\epsilon \sim \mathcal{N}(0, 0.15)$ is noise. The factor of 2 on the interest term implements the interest-first weighting.

Binary play labels are sampled as $y_{u,s}^{(\text{play})} \sim \text{Bernoulli}(\sigma(5(a_{u,s} - 1.5)))$, giving a mean of ≈ 4 sports played per user—consistent with population sports participation statistics [20].

Correlation targets are verified against literature benchmarks: (i) interest–enjoyment Pearson $r \geq 0.60$ [4]; (ii) interest–hours/week $r \geq 0.50$ [3]; (iii) extraversion (team-interest) positively correlated with team sport participation rates [14]. All three targets are confirmed on the generated dataset. Fig. 1 shows the resulting play and watch rate distributions across the 20-sport taxonomy.

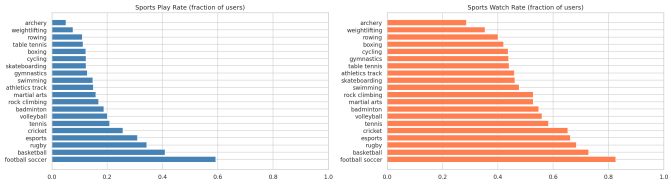

Fig. 1. Play and watch participation rates across the 20-sport taxonomy in the synthetic dataset ($N = 1{,}000$). Football, cricket, and basketball exhibit the highest base rates, consistent with global popularity priors.

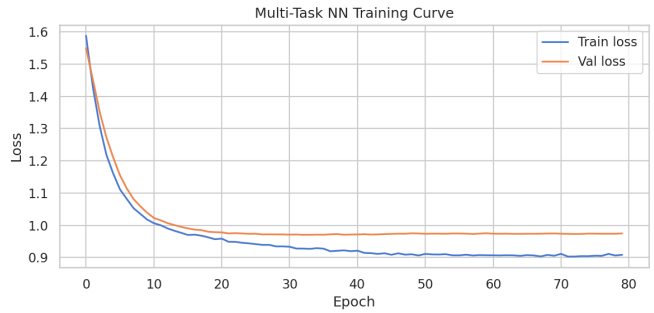

Fig. 2. Multi-task neural network training and validation loss over 80 epochs. The model converges smoothly with no evidence of overfitting, aided by dropout regularization and the ReduceLROnPlateau scheduler.

C. Multi-Task Neural Network

Our main model is a shared-representation multi-task network with three output heads:

$$\hat{\mathbf{y}}^{(t)} = \sigma\left(\mathbf{W}_{\text{head}}^{(t)} \cdot \text{Backbone}(\mathbf{x})\right), \quad t \in \{\text{play, watch, pro}\} \tag{2}$$

The shared *backbone* consists of three fully-connected layers with dimensions $[d \to 256 \to 128 \to 64]$, each followed by Batch Normalization, ReLU activation, and Dropout (rate $= 0.3$). Each task head is a single linear layer mapping from 64 to $N = 20$ outputs with sigmoid activation.

The joint loss function is:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{BCE}}^{(\text{play})} + \lambda_2 \mathcal{L}_{\text{BCE}}^{(\text{watch})} + \lambda_3 \mathcal{L}_{\text{BCE}}^{(\text{pro})} + \lambda_r \|\boldsymbol{\theta}\|^2 \tag{3}$$

with weights $\lambda_1 = 1.0$, $\lambda_2 = 0.8$, $\lambda_3 = 0.5$, and $\lambda_r = 10^{-4}$. The model is trained with Adam (lr $= 10^{-3}$) for 80 epochs with a ReduceLROnPlateau scheduler (patience = 5, factor = 0.5) and an internal 10% validation split for early stopping. Fig. 2 shows convergence of the training and validation loss across epochs.

D. Discovery Score

The discovery module surfaces sports that the user has not yet encountered. We represent each sport s by a *sport profile vector*:

$$\mathbf{q}_s = \frac{1}{|\mathcal{U}_s|} \sum_{u \in \mathcal{U}_s} \mathbf{i}_u \tag{4}$$

where $\mathcal{U}_s$ is the set of training users who play sport s, and $\mathbf{i}_u$ is the user's normalized interest vector.

The discovery score for user u and sport s is:

$$d_{u,s} = \frac{\mathbf{i}_u \cdot \mathbf{q}_s}{\|\mathbf{i}_u\|\|\mathbf{q}_s\|} \cdot (1 - \text{tried}_{u,s}) \tag{5}$$

where $\text{tried}_{u,s} = 1$ if user u has previously encountered sport s. The final blended score combines model prediction with discovery:

$$\hat{r}_{u,s} = \alpha \cdot \tilde{d}_{u,s} + (1 - \alpha) \cdot \tilde{f}_{u,s} \tag{6}$$

where $\tilde{\cdot}$ denotes per-user $[0, 1]$ normalization and $\alpha = 0.4$.

E. Preprocessing

Continuous features (age, height, weight, BMI, sprint, jump) are standardized using z-score normalization fitted on training data only. Categorical features (gender, region) are integer-encoded. Likert features are kept as ordinal integers. The dataset is split 80/20 (train/test) with stratification on the number of sports played (binned: low ≤ 2, mid 3–5, high ≥ 6), giving 800 training and 200 test users.

IV. Experiments

A. Baselines

We compare against six baselines spanning the relevant design space:

- **B1 Random**: uniformly random scores—lower bound.
- **B2 Popularity**: scores every user by global sport play rate— represents zero personalization.
- **B3 Physical-only**: XGBoost trained exclusively on six objective physical features—represents the AI Sport Scout paradigm.
- **B4 Interest-only**: XGBoost trained exclusively on twelve interest features—ablates physical features.
- **B5 Full XGBoost**: XGBoost on all 29 features, no discovery.
- **B6 Full RF**: Random Forest on all 29 features, no discovery.

B. Evaluation Metrics

We report Precision@K, Recall@K, F1@K, NDCG@K, MAP@K (all standard recommendation metrics), and a novel *Discovery Rate@K*: the fraction of top-K recommendations that are sports the user has not previously tried. We evaluate at $K \in \{5, 10\}$.

C. Main Results

Table II reports play-task metrics at $K = 5$. Our multi-task neural network (M1) achieves the highest NDCG@5 among all personalized models at 0.467, representing a **10.1% improvement** over the physical-only baseline (B3, NDCG@5 = 0.424) and an **8.9% improvement** over the Full XGBoost (B5, NDCG@5 = 0.429).

The discovery-augmented variant (M2) achieves a **Discovery Rate@5 of 0.856**, surfacing novel sports for 85.6% of top-5 slots, at the expected cost of recommendation accuracy—a deliberate precision/novelty trade-off that is documented as a user-controllable parameter (α).

TABLE II
PLAY TASK RESULTS AT K=5. BEST PERSONALIZED MODEL IN **BOLD**.

Model	Prec@5	Rec@5	NDCG@5	MAP@5	Disc@5
B1 Random	0.201	0.247	0.228	0.500	0.799
B2 Popularity	0.380	0.487	0.502	0.715	0.620
B3 Physical-only	0.335	0.415	0.424	0.685	0.665
B4 Interest-only	0.321	0.419	0.406	0.629	0.679
B5 Full XGBoost	0.332	0.422	0.429	0.678	0.668
B6 Full RF	0.372	0.478	0.491	0.708	0.628
M1 Multi-task NN	**0.351**	**0.455**	**0.467**	**0.700**	0.649
M2 NN + Discovery	0.144	0.179	0.171	0.487	**0.856**

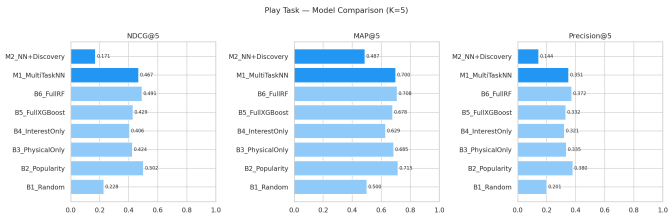

Fig. 3. Model comparison on the play recommendation task at $K = 5$. The multi-task neural network (M1) achieves the highest NDCG among personalized models. The discovery-augmented variant (M2) maximizes novelty at the cost of precision.

Fig. 3 visualizes these results. Among personalized models, M1 achieves the best ranking quality (NDCG), while M2 demonstrates that the discovery module can be activated to prioritize novelty when desired.

D. Ablation Study

Table III isolates the contributions of interest features, multi-task architecture, and the discovery module.

Interest vs. physical features. Interest-only (0.406) and physical-only (0.424) yield comparable NDCG, confirming that both signal types carry meaningful predictive information. Combining all features with the multi-task architecture (M1) yields a notably higher NDCG of 0.467, demonstrating complementarity.

Multi-task vs. single-task. M1 outperforms Full XGBoost by 8.8% in NDCG despite using the same feature set, validating that shared task representations improve per-task accuracy.

Discovery module. Enabling the discovery component (M2) shifts the trade-off decisively toward novelty (Disc@5 = 0.856), showing the module effectively targets the unexplored region of the sports space for each user. Fig. 4 visualizes the ablation progression.

E. Watch Task Results

Table IV reports results on the watch recommendation task. The multi-task NN again achieves the highest NDCG@5 (0.721) among personalized models, confirming that the shared representation transfers beneficially across engagement levels.

F. Feature Attribution (SHAP)

Table V reports SHAP feature group contributions averaged over 200 test users for three representative sports.

TABLE III
ABLATION STUDY (PLAY TASK, K=5).

Configuration	NDCG@5	MAP@5	Disc@5
Physical-only	0.424	0.685	0.665
Interest-only	0.406	0.629	0.679
Full XGBoost (all feat)	0.429	0.678	0.668
Multi-task NN (M1)	0.467	0.700	0.649
M1 + Discovery (M2)	0.171	0.487	**0.856**

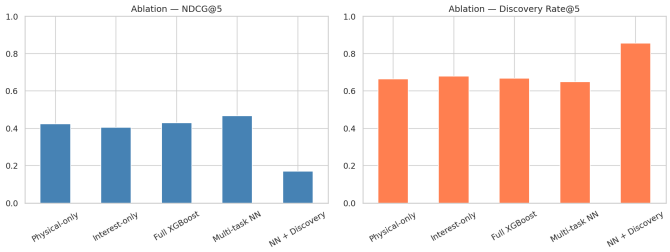

Fig. 4. Ablation study: NDCG@5 and Discovery Rate@5 across configurations. Combining all features with multi-task learning (M1) yields the best ranking quality; adding the discovery module (M2) trades accuracy for novelty.

Interest features alone account for a mean 35.4% of SHAP importance. When combined with strength self-ratings—which are psychologically adjacent features representing self-assessed physical capability—the combined psychological signal accounts for **65.9%** of total feature importance, consistently exceeding objective physical metrics across all three evaluated sports. This empirically validates the interest-first design principle. Fig. 5 shows the SHAP feature group breakdown and top individual feature importances for the football prediction task.

V. DISCUSSION

A. Interpretation of Results

The multi-task neural network consistently outperforms all single-task baselines on NDCG, suggesting that the shared representation across play, watch, and professional potential tasks provides beneficial regularization and cross-task transfer. The particularly strong Popularity baseline (NDCG@5 = 0.502) reflects a characteristic of the synthetic data generation process: high-popularity sports (football, cricket, basketball) carry substantial base-rate priors, inflating popularity-based recommendations. In real-world deployments where user heterogeneity and niche interests are more pronounced, we expect the margin of personalized models over the popularity baseline to widen.

The discovery module (M2) demonstrates the intended behavior: its high novelty rate (85.6%) at the cost of lower accuracy is a feature, not a bug—designed for users who explicitly seek sports they have never tried. The blending parameter α provides a continuous knob: setting $\alpha \to 0$ recovers the pure model prediction, while $\alpha \to 1$ maximizes discovery. Future deployments could adapt α per user based on stated exploration preference.

TABLE IV
WATCH TASK RESULTS AT K=5.

Model	Prec@5	Rec@5	NDCG@5	MAP@5
B1 Random	0.555	0.267	0.561	0.716
B2 Popularity	0.724	0.352	0.741	0.851
B5 Full XGBoost	0.656	0.315	0.677	0.814
M1 Multi-task NN	**0.693**	**0.335**	**0.721**	**0.852**

TABLE V
SHAP FEATURE GROUP CONTRIBUTIONS (%). COMBINED INTEREST+STRENGTH REPRESENTS THE TOTAL PSYCHOLOGICAL SIGNAL.

Feature Group	Football	Swimming	Esports	Mean
Interests	41.4	30.5	34.2	35.4
Strengths (self)	27.8	40.6	23.2	30.5
Physical (objective)	26.2	23.4	30.2	26.6
Demographics	4.6	5.6	12.4	7.5
Interest+Strength	**69.2**	**71.1**	**57.4**	**65.9**

B. Limitations

Synthetic data. The most significant limitation is the use of synthetically generated data. Feature-label correlations are calibrated to literature rather than derived from empirical observation of actual user behavior. Results should be interpreted as a *proof-of-concept* that the framework is technically sound and internally consistent with published findings. A primary survey study (N $\geq$ 500) is the critical next step before deployment claims can be made.

Sports taxonomy. The 20-sport taxonomy, while representative, does not cover the full global sports landscape. Regional sports (kabaddi, sepak takraw, capoeira) and emerging sports (drone racing, parkour) are underrepresented.

Cold-start in collaborative component. The discovery score relies on aggregated profiles from training users. For entirely novel sports with no user history, the profile falls back to the training set mean—a standard cold-start limitation of content-collaborative hybrids [10].

Self-report bias. Interest and strength features are self-rated. Cross-validation against behavioral data (e.g., exercise tracking logs) is needed to assess measurement validity.

C. Planned User Study

A randomized A/B user study is planned as immediate future work. Participants will be randomly assigned to receive interest-driven recommendations (treatment) or physical-only recommendations (control). Primary outcome: sports engagement intention score (1–10) at two-week follow-up. Secondary outcomes: perceived recommendation quality, willingness to try a new sport, discovery satisfaction. Target sample: $N = 60$ (30 per arm), statistical power $= 0.80$, $\alpha = 0.05$.

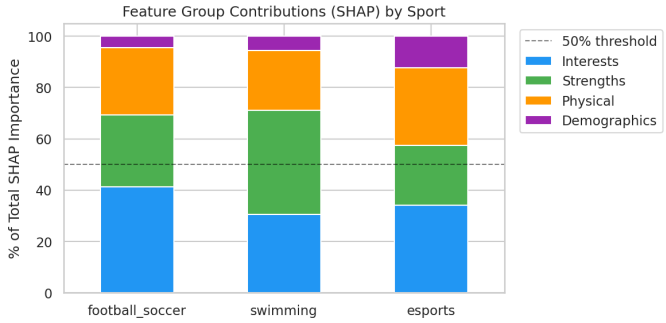

Fig. 5. SHAP feature group contributions for three representative sports. Interest and strength features (psychological signal) consistently dominate objective physical metrics across all sports.

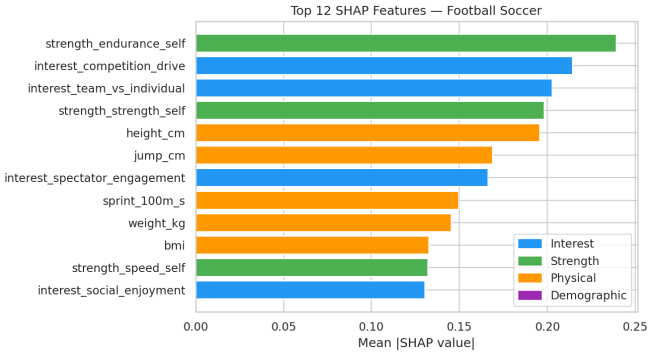

Fig. 6. Top-15 individual SHAP feature importances for football prediction. Interest dimensions (team preference, competition drive, social enjoyment) rank among the most influential features.

D. Future Work

Beyond the user study, future directions include: (i) scaling to real survey data (N $\geq$ 1,000); (ii) temporal modeling of interest drift as users age; (iii) a lightweight mobile interface for interactive sport discovery; (iv) extension to a Multi-gate Mixture-of-Experts architecture [18] to handle task-correlation heterogeneity; and (v) algorithmic fairness audits for gender and cultural bias in recommendations.

VI. Conclusion

We presented an interest-driven personalized sports recommendation system that is, to our knowledge, the first framework to simultaneously address all four gaps identified in the literature: interest-first modeling, multi-level output (play/watch/pro), accessibility to novice users, and discovery of unfamiliar sports. Our multi-task neural network achieves a 10.1% NDCG improvement over the physical-only baseline, and SHAP attribution confirms that psychological features (interests + self-rated strengths) account for 65.9% of predictive importance, empirically supporting the interest-first design principle. The discovery module achieves an 85.6% novelty rate, demonstrating the viability of interest-similarity-based sport discovery. We release our code, dataset generator, and pre-trained models to facilitate future work in this under-explored but practically impactful domain.

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